

CSIT606

Enrol. No.

[ET]

END SEMESTER EXAMINATION : APRIL-MAY, 2023

INTRODUCTION TO DATA ANALYTICS

Time : 3 Hrs.

Maximum Marks : 50

Note: Attempt questions from all sections as directed.

SECTION – A (20 Marks)

*Attempt any **four** questions out of five.*

*Each question carries **05** marks.*

1. How Big Data Analytics can be useful in the development of smart cities. (Discuss one application)
2. Which method of data analysis should I use for a study using questionnaires?
3. How would policy iteration be defined for action values?

4. Why data cleansing is important in data analysis?
5. List the differences between overfitting and underfitting?

SECTION - B (16 Marks)

*Attempt any **two** questions out of **three**.*

*Each question carries **08** marks.*

6. Hospital records show that 75% of patients suffering from a disease die due to that disease. What is the probability that 4 out of 6 randomly selected patients recover?
7. Write explanatory notes on
 - (a) Q Learning
 - (b) Actor-Critic
8. (a) Differentiate between Numerical Data and Categorical Data.

- (b) Differentiate between skewed and uniform distribution?

SECTION – C (14 Marks)
(Compulsory)

9. Cluster the following eight points (with (x, y) representing locations) into three clusters:

A1(4, 20), A2(4, 10), A3(8, 4), A4(10, 8), A5(7, 10),
A6(6, 4), A7(1, 4), A8(4, 9)

Initial cluster centers are: A1(4, 20), A4(10, 8) and A7(1, 4).

The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is defined as-

$$P(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

Use K-Means Algorithm to find the three cluster centers after the second iteration.